

2 THINKING LIKE AN ECONOMIST

INTRODUCTION

A perpetual debate in many economies revolves around the provision of health services. In the UK and many European countries, health provision is universal, meaning it is available to all regardless of their income or status and is free at the point of use. Of course, health provision is not 'free' – it is extremely expensive. If a government wants to increase spending on health, it must find a way of funding it. Assume that a government announces an increase in spending on its health service of €2.8 billion, to be funded by new measures to prevent large corporations avoiding their tax liabilities by tightening the rules on corporate taxes.

How would an economist think about this policy? An economist would want to know what the additional investment would be spent on, whether this additional spending would result in a more efficient health service and, crucially, would want to know how 'efficient' was being defined in this context.

Regardless of whether the economist has a personal view about whether the means of raising the funds are 'right', they would think about whether the amount of money raised through these measures would be sufficient, and whether tightening rules on tax avoidance would have consequences on the behaviour of economic actors who would be affected. Political parties might not foresee these changes in behaviour, and this could compromise the intended outcome.

Ultimately, the economist would want to investigate the costs and benefits of such a policy, try to quantify those costs and benefits, and offer an informed view of the consequences. It would not simply be a case of looking at the obvious costs and benefits but also the hidden costs and benefits which might lead to an outcome or outcomes that are very different from those the policy was designed to achieve.

Economics, like most other fields of study, has its own language, its own processes, its methods of discovery and its own way of thinking. As you embark on your study of economics you will have to learn lots of terms and concepts. Many of the concepts you will come across in this book are abstract. Abstract concepts are ones which are not concrete or real – they have no tangible qualities. We will talk about markets, efficiency, comparative advantage and equilibrium, for example, but it is not possible to physically see these concepts.

As you work through your modules you will find that it is not always easy to think like an economist, and there will be times when you are confused and find some of the ideas and concepts being presented to you running contrary to common sense (i.e. they are counter-intuitive). What you will be experiencing is perfectly normal and a part of the learning journey.

ECONOMIC METHODOLOGY

How do economists know what they know? What methods do they adopt to find out information and arrive at theories? In this chapter we will discuss the methodology of economics. There is considerable debate about this methodology and, crucially, about the assumptions which underpin the discipline. There has been, and probably will continue to be, a number of books and articles published which are critical of economics and economists. This has been exacerbated by the Financial Crisis of 2007–9. If you read

some of these books and articles, you might be forgiven for wondering what on earth you have done in choosing to study such a bankrupt subject which is populated by automatons who ignore everything 'real' and blindly pursue their own narrow focus because it suits them to do so.

The reality is somewhat different. There have been debates and disagreements in economics for hundreds of years; some of these disagreements are about the assumptions that are made in exploring economic phenomena. In other cases, there may be broad agreement on the direction of (for example) cause and effect, but the disagreement may be on the extent of the effects.

Economics as a Science

One of the debates about economics is the extent to which it is a 'science'. Science is a process; it is related to the discovery and creation of new knowledge and understanding but also relies on existing knowledge and understanding. Science is ongoing. The knowledge and understanding associated with the process are constantly evolving as new discoveries help improve our knowledge and understanding of the world around us.

Of course, we tend to think of science from the perspective of physics, chemistry and biology, which many people have studied at school. These subjects are referred to as 'natural sciences', because they are associated with the study of physical things and the natural world. When studying natural phenomena, it is often possible to conduct controlled experiments. This means that researchers can vary an object of interest and observe what happens to other variables and objects. The experiment can be repeated, and data gathered, which can help in the explanation of events and to establish cause and effect.

Other discipline areas cannot carry out experiments in the same way. Economics is one of those disciplines. Economics studies decision-making and the effect of decision-making on a wide range of topic areas, but central to the study is human beings. Controlled experiments which can be carried out in the natural sciences cannot be carried out in the same way in economics. Economics is referred to as a 'social science' because it deals with human beings as individuals and in groups. The process of knowledge creation and development in social sciences can take on different nuances compared to the natural sciences, but there are processes and methods which are common to both.

SELF TEST Can any discipline which deals with human behaviour truly call itself a 'science'?

Models

Economics uses a lot of models. A model is a representation of reality which facilitates understanding of how something works. Models can be used as a means of helping understand the real world and for making informed decisions and judgements.

Models are, of necessity, simplifications of reality and not meant to represent every feature, nuance or aspect of the real world it is attempting to explain. It is often worth thinking of models which architects use to show how a building will look. The model will provide the observer with an image of what the eventual building will look like. It shows its key features and helps in understanding the scale of the building, how it integrates with its surroundings and its main structures. What the model does not do is incorporate every feature and aspect of the building – that is not necessary to develop a broad understanding of the building and its environment.

Similarly, economists use models to represent the world around them. We use models to represent how markets work, how the economy as a whole works, how consumers behave and how firms behave. These models are based on assumptions, some of which might not be fully accurate as a representation of how the real world works or how the economic agents which form part of the model behave. This does not necessarily detract from the value of the model in describing how the phenomenon under investigation works.

Economic models have two principal uses: one is for predicting or forecasting what might happen in the future as a consequence of a decision or policy, and the other is to simulate an event and provide

a comparison with what would have happened if the decision, policy or change had not happened (the **counterfactual**). Economists' models are most often composed of diagrams and equations. By feeding in data, economists can use models to generate outcomes which provide some insight and form the basis of decision-making.

counterfactual analysis is based on a premise of what would have occurred if something had not happened

Models are valuable in that they allow economists to manipulate variables which form part of the model and explore what might happen. Economics models will always contain a number of variables. Some of these variables are determined by the model and some are generated within the model. For example, take the market model where the quantity demanded (Q_d) is dependent on the price (P). Q_d is said to be the dependent variable. Its value will be dependent on the functional relationships in the model (the factors that affect demand) such as incomes, tastes and the prices of other goods. Q_d can be described as an **endogenous variable**. Price, on the other hand, is the independent variable – it affects the model (the quantity demanded) but is not affected by it. The price is not determined by, or dependent on, the quantity demanded. Price would be referred to as an **exogenous variable**.

endogenous variable a variable whose value is determined within the model

exogenous variable a variable whose value is determined outside the model

Models are inherently unstable the longer the time period being considered and forecast. Shocks occur which are impossible to factor into the building of models. These not only have short-term impacts but may also change longer-term dynamics. For example, the attacks on the World Trade Centre on 11 September 2001 have had a fundamental impact on the ways in which governments think and behave that could not have been envisaged before the event. One of the reasons why models of climate change are subject to debate and disagreement is that over time the internal dynamics of models change in ways which render future predictions inherently unstable.

The so-called 'butterfly wing' effect, as described in chaos theory, highlights the complexity surrounding modelling in meteorology. The butterfly effect notes that a butterfly flapping its wings at a particular point in time and space creates small changes in conditions which can lead to significant changes in faraway places, such that a flap of a butterfly's wings in New Mexico could be traced as the initial causal factor of a hurricane in China sometime in the future. Chaos theory further tells us that minor errors in measurements or assumptions can be amplified to such an extent that any predictions made by the model are rendered useless, and that the further into the future we are attempting to make forecasts and predictions, the more unstable our models are.

Cause and Effect One problem facing economists is separating out cause and effect. Observation and experience can lead to the identification of phenomena occurring which intuition would seem to suggest are related in some way. Does a change in price, for example, cause a change in the amount bought by consumers, or does quantity bought affect price?

To get a clearer picture, economists will utilize an important aspect which is common to other sciences, that of holding other variables in the model constant. The Latin term '**ceteris paribus**' meaning 'other things equal', is used to note when other factors that might affect outcomes are assumed to be constant. Research can lead to a conclusion which provides an answer. The question which must be asked is 'How do we know this "answer" is correct?'

ceteris paribus (other things equal) a term used to describe analysis where one variable in the model is allowed to vary while others are held constant

Take the case of the relationship between crime rates and unemployment. Is a rise in the crime rate, for example, *always* caused by a rise in unemployment, or are there other factors that can also influence the crime rate? How significant is the incidence of unemployment in determining crime rates? How do those who research such a relationship establish the facts of the case?

Human Values in Models Models can be developed, predictions made and conclusions drawn, but there are then human values to take into consideration. Many economists, for example, will agree that there is sufficient evidence to suggest that government stimulus in a period of economic downturn can help reduce the number of people unemployed. What might be the subject of more disagreement is the significance of the effect, or the value of the associated costs and benefits of such a policy.

Models of climate change may suggest that the increase in human-generated carbon emissions will contribute to a change in the global climate. There may be some who would disagree with this basic conclusion, partly because they dispute the ‘facts’ which form the basis for the model.

Models also allow **inferences** to be made. This means that conclusions, consequences or explanations can be drawn based on the evidence provided by the model. This is not to say that these conclusions are full and final; they are simply what may be reasonably and logically derived based on the manipulations of the model.

inference a conclusion or explanation derived from evidence and reasoning

The climate change model, for example, might infer a policy suggestion that significant measures might have to be put in place to cut carbon emissions in the next 10 years to prevent the costs which our children and grandchildren will have to bear. There may be people who disagree on whether the cost of the current sacrifices required are outweighed by the value of the benefits that will occur between 50 and 100 years into the future.

Manipulating Models Economists will often use models based on mathematical formulae. This can allow the modeller the ability to manipulate the numbers in the formula and identify the extent to which outcomes differ. When a model is manipulated, outcomes can be identified. The model may help to explain the mechanism or reasons why the outcomes identified occur. The outcomes from models can then be compared to actual data to see the extent to which the model is useful in explaining observed data and behaviour.

This is a perfectly normal part of the scientific process. Those critical of a model and its outcomes can provide refinements to the model which might better represent the phenomena it is meant to be describing or explaining. This is how knowledge is built, developed and improved upon. The explanatory power of models is dependent on how well they are built. If they are too simplified or the assumptions cannot be reasonably observed in the real world, then their explanatory power breaks down.

SELF TEST Make a list of five benefits of modelling in economics and five limitations.

Types of Reasoning

One of the ways in which science discovers new knowledge is through asking questions. The consequences which arise from asking these questions can be significant. For example, if Isaac Newton really did get hit on the head with an apple and, amidst his pain, asked the question ‘I wonder why apples fall to the ground,’ the answers he generated have fundamentally changed the way we look at the world. Newton’s work on gravity spawned many other questions and led Einstein to arrive at the theory of relativity, and the theory of relativity was used to help in the development of global positioning satellites (GPS) which so many people in the world now use and rely upon in their cars, smartphones, watches and other gadgets.

When questions are asked, there are different routes which scientists take to explore those questions or, in some cases, arrive at the questions themselves. We can identify different types of reasoning which help clarify the process involved. There is no 'right' way of reasoning, but there is debate about which produces more reliable theories, which in turn have predictive power.

Deductive Reasoning Deductive reasoning begins with known 'facts' or 'truths' – things that we know to be true (or think to be true). It then works through a process of using these facts or truths to arrive at answers to the question we are interested in and, as a consequence, arriving at new facts or truths.

The 'question' might take the form of a general statement or **hypothesis**. The word is derived from the Greek (*hypotithenai*) meaning 'a placing under' or 'to suppose'. A hypothesis is an assumption, a tentative prediction, explanation, or supposition for something. To discover whether the hypothesis is true or correct, it must be tested. If the facts or known truths are applied to the hypothesis, then the conclusions drawn allow us to discover whether the hypothesis is 'true' or 'correct'.

hypothesis an assumption, tentative prediction, explanation, or supposition for something

A very simple example can serve to highlight deductive reasoning. You observe an animal you haven't seen before and ask the question 'I wonder if that animal is a bird?' The hypothesis or supposition would be that this animal is a bird. You observe that the animal has feathers. Based on the known truth that all birds have feathers, if this animal has feathers then you can conclude that it is a bird.

The conclusion rests on the assumption that the facts or known truths used to arrive at the conclusion are indeed true, in this case, that all birds have feathers. If the facts or the premise used in our example was 'All birds have feathers and fly,' we might arrive at a conclusion which is unsound. This is because while all birds have feathers, not all birds fly. This simple example serves to highlight one of the reasons why there can be disagreement in economics. The 'truths' or facts that are used in deductive reasoning might be disputed. We will see this highlighted in our discussion on consumer behaviour later in the book. The assumptions or 'truths' that are used to explain human behaviour and arrive at conclusions in this context have been disputed and given rise to alternative conclusions.

Let us look at a simple example in economics. Many countries have put in place legislation to establish a minimum wage as a means of protecting the lowest paid workers in society. What will be the consequences of this legislation? An economist might develop a hypothesis that a minimum wage will result in increased unemployment. In analyzing whether this hypothesis is 'true' or not, the economist might use 'known facts' that when the price of something is set above equilibrium, the quantity demanded will fall and the quantity supplied will rise, resulting in a surplus. In this example, the surplus will be a surplus of labour, i.e. more people willing to supply labour services compared to the demand by employers of those labour services. Therefore, the conclusion is that a minimum wage will indeed result in unemployment.

Economists may use models of the labour market which are based on mathematical equations to help quantify the extent of the unemployment which will occur. These models are in turn based on the 'known truths' about the labour market and how it works. There is extensive debate around minimum wage laws, which is a particular source of disagreement among economists. Part of the reason for this is disputes over the 'known truths' and the parameters of the model, the way the variables are defined and quantified. This is one of the reasons that economics differs from the natural sciences, in that the models used are based on human behaviour and not on natural forces. Human behaviour tends to be unpredictable and not always fully understood, whereas factors in natural science may be more stable and 'easier' to define and quantify.

Inductive Reasoning Inductive reasoning begins with data and observation. The data or observations are analyzed. From this analysis, patterns are identified, which may be patterns of behaviour. These patterns generate a question, or hypothesis, which explains the observed behaviour or pattern. This explanation or conclusion is then applied to all other instances of the phenomena. This is referred to as '**generalization**'. In generalizing, the researcher is offering a theory or explanation of events and phenomena. This theory

can then be tested and verified, or could be shown to be inaccurate and amended, or a new theory proposed to replace the incorrect one.

generalization the act of formulating general concepts or explanations by inferring from specific instances of an event or behaviour

One example of inductive reasoning in economics might be the observation that countries experiencing rapid and accelerating inflation (hyperinflation) also seem to have central banks which print large sums of money. The 'pattern' seems to be that rapid increases in the money supply are associated with instances of hyperinflation. If there was only one instance of this 'pattern', it may be that the researcher could not generalize to all instances, but if this pattern was observed and verified by the data in a number of instances, it may be possible to arrive at a general theory which posits that hyperinflation is caused by rapid increases in the money supply. Other instances of hyperinflation can then be monitored, the data analyzed, and the theory could be confirmed or rejected, depending on the nature of the evidence which arises from additional instances of this phenomena.

Inductive reasoning is thus empirical in nature – it refers to evidence to confirm or reject theories. For this reason, inductive reasoning is seen as being a benchmark for developing knowledge and understanding. One of the challenges facing social sciences like economics in using inductive reasoning is that conclusions and generalizations can be made based on partial or incomplete data.

Given the nature of human beings, this is a particular problem for economics. If we observe patterns of behaviour for tens of thousands of households in Ireland, for example, and draw a general conclusion and theory of this behaviour, can we safely assume that this also explains similar behaviour across Europe or indeed the world? Identifying patterns in data means that the data must be collected, be available for study and be reliably complete. This is not always possible. Looking at data for gross domestic product for different countries online, for example, often highlights differences in results from one source to another. This may be explained by the way in which the data is collected, how it is processed, who is gathering and analyzing the data, and the statistical processes used.

Experiments in Economics

Although economists use inductive reasoning like other scientists, they do face an obstacle that makes their task especially challenging. Physicists, for example, can set up controlled experiments such as the Large Hadron Collider, which is seeking to recreate conditions that existed milliseconds after the Big Bang. The experiments being conducted are designed to help confirm existing theories and/or develop new ones to explain forces and matter and how the universe began. By contrast, economists studying inflation are not allowed to manipulate a nation's monetary policy simply to generate useful data.

Economists pay close attention to the natural experiments offered by history. When political instability interrupts the flow of crude oil, for instance, oil prices rise around the world. For consumers of oil and oil products, such an event depresses living standards. For economic policymakers, it poses a difficult choice about how best to respond. For economists, it provides an opportunity to study the effects of a key natural resource on the world's economies, and this opportunity persists long after the increase in oil prices is over.

Throughout this book, therefore, we consider many historical episodes. These episodes are valuable to study because they give us insight into the economy of the past and, more importantly, they allow us to illustrate and evaluate economic theories of the present.

Despite the challenges faced by economists in conducting experiments, there are two major fields that are worthy of note. Experiments in economics can be conducted in a 'laboratory', where data can be collected via observations on individual or group behaviour through questionnaires and surveys, interviews and so on, or through the collection and analysis of data that exists such as wages, prices, stock prices, volumes of trades, unemployment levels, inflation and so on. The data can be analyzed in relation to a research question and conclusions drawn which help develop new understanding or refine and improve

existing understanding. The conclusions drawn from such experiments may be generalizable; in other words, the findings of the experiment can be extended outside the 'laboratory' to explain behaviour or economic phenomena and provide the basis for prediction.

One example of how such laboratory experiments can help change understanding is the work of people such as Daniel Kahneman, Amos Tversky, Richard Thaler and Cass Sunstein, whose research has helped to provide insights into judgement and decision-making and has offered a different perspective on the assumptions of rational decision-making. Thaler, the 2017 Nobel Prize winner, for example, conducted a number of experiments to explore how individuals respond when faced with different questions on losses and gains in relation to a reference point. He found that prior ownership of a good, for example a ticket to see a football game, altered people's willingness to sell, even at prices significantly higher than they had paid.

Thaler's observations on the consistency of this behaviour across a number of experiments led him to coin the term '*endowment effect*' to explain the behaviour. It is now widely accepted that the endowment effect does exist and that it runs counter to the assumptions of rational behaviour in economics. Thaler worked with Kahneman and Tversky and extended the theory to distinguish between goods which are held for trade and those which are held for use. The endowment effect, they suggested, was not universal; it was more powerful when goods were held for use.

A second type of experiment in economics is *natural experiments*. A natural experiment is one where the study of phenomena is determined by natural conditions which are not in the control of the experimenter. Natural experiments can be exploited when some change occurs which allows observation to be carried out on the effects of this change in one population, and comparisons made with another population that is not affected. Examples of natural experiments include observing the effects of bans on smoking in public places on the number of people smoking or the possible health benefits; how far a change in the way in which education is financed affects standards; the effect on income of the years spent in education; the effects of a rise in a tax on property on the market for housing; and the effects on the female labour market of changes in fertility treatment and availability.

Typically, natural experiments make use of the statistical tools of correlation and regression to determine whether there is any relationship between two or more variables; if any such relationship exists; and, if it does, what the nature and strength of the relationship is. From such analysis, a model can be developed which can be used to predict.

At the heart of such analysis is the extent to which a relationship between two or more variables can be linked to cause and effect. Just because two variables appear to have some relationship does not necessarily imply cause and effect. For example, a researcher might find that an observation of graduates in the workforce shows that their incomes are generally higher than those of non-graduates. Can the researcher conclude that having a degree will lead to higher income? Possibly, but not necessarily. There might be other factors that have an effect on income apart from having a degree. Trying to build a model which takes into account these different factors is an important part of the value of natural experiments.

CASE STUDY The Basic Income Experiment

It is sometimes said that economists cannot conduct major experiments by intervening in economies, but in Finland they have done just that. A two-year pilot was introduced in January 2017 to investigate the effect of the introduction of a basic income scheme. Some 2,000 participants were chosen at random to take part in the experiment.

The hypothesis being investigated was whether giving the unemployed a guaranteed basic income had an impact on their employment prospects. The income amount was the same for every individual regardless of their background or position and was paid periodically, for example, every month. Unlike many unemployment insurance schemes, receiving basic income was not dependent on the individuals having to demonstrate that they were seeking employment. If an individual received a guaranteed basic income, would this help alleviate poverty and inequality, and encourage individuals to find work?

(Continued)

A number of similar experiments are in operation related to basic income. In Canada, for example, a basic income experiment was introduced in June 2017 in two cities and a town in Ontario. Two groups of people were selected. One group received a basic annual income of up to CDN\$17,000 (around €11,100) and the other group received nothing. In the Canada experiment, the participants must be on a low income and, for those getting the basic income, if they find work their basic income is reduced by half.

Barcelona launched a basic income experiment in October 2017 which, like the Finland experiment, involved 2,000 participants, half of whom received between €400 and €525 per month over a two-year period. Unlike the Finland experiment, in Barcelona the money was given to households, not individuals, and those receiving the money were required to give something back in the form of attending support programmes to help them find work, as well as other community-based programmes.

In Finland, the 2,000 unemployed participants received €560 per month as a basic income. By having this guaranteed sum, the hypothesis was that individuals would be likely to be more flexible in seeking work, moving between jobs and taking part in the gig economy with all its potential uncertainties. The experiment was due to last two years and an evaluation of the results of the experiment was considered along with other welfare measures in the country.

The experiment was run from the Social Insurance Institution, a government agency. The Institution applied for an extension of the experiment to include individuals who were employed, but the request was turned down by the Finnish government in April 2018. The experiment on the initial 2,000 participants continued as planned and the results were analyzed and published in the latter part of 2019.



If an individual receives a guaranteed basic income, would this help alleviate poverty and inequality, and encourage individuals to find work?

Theories

Theories can be used to explain something and to make predictions. The theory of indifference curves and budget lines can be used to explain consumer behaviour. The value of this theory is how reliable it is in predicting consumer behaviour. If we observe the way consumers behave and the outcomes predicted do not conform to the theory, it may be that new research must be conducted to offer refinements to the theory, or even consigning the theory to history.

One of the criticisms of economics is that some theories commonly taught on undergraduate courses have been derived through deductive reasoning, but that the premises used as the basis for the conclusions drawn are inaccurate, or just wrong, and not supported by data and evidence. For example, the neo-classical theory of consumer behaviour makes assumptions that consumers act rationally, prefer more to less and make purchasing decisions based on pure self-interest. The premises for the theory were developed in the nineteenth century when the economy and society were very different. The historical context of many theories in economics should not be ignored.

Critics argue that these assumptions are not supported by evidence and data, and thus any predictions arising from such theories are unreliable or wrong. Supporters of such theories argue that they do contain useful insights into behaviour that allow valid predictions to be made. While they may not explain human

behaviour in every instance, they still have some value. Models, they argue, are after all simplifications of reality and cannot hope to replicate every aspect of human behaviour.

Falsifiability

As hinted at earlier, one of the criticisms of economics in recent years has been that while it claims to be a science, it does not follow scientific principles, or at least only does so when it deems it convenient. In particular, critics have argued that many of the theories we will present in this book are no longer accurate and should be dismissed. Despite this, and critics would argue that this book is one of many that perpetuate the problem, these theories continue to form the basis for many undergraduate economics courses. Why do we continue to teach theories which are inaccurate or just wrong? Critics would argue that including these theories in an economics course is like physics courses continuing to teach a theory of the earth being flat.

In considering these debates, we can refer to the philosophy of science and one of its foremost exponents, Sir Karl Popper. Popper was born in Vienna in 1902 and in 1946 moved to the UK to teach at the London School of Economics. He was knighted in 1965, and his contribution to the philosophy of science was extensive and highly regarded.

One of Popper's important contributions was the principle of **falsifiability**. Popper based the idea on the basic assumption that it is not possible to know the truth of everything. Ideas and theories might be widely accepted and adopted, but ultimately, we cannot be 100 per cent sure that these ideas and theories are correct. Knowledge is always subject to evolution and development, and in the light of new evidence, our theories and ideas may change.

falsifiability the possibility of a theory being rejected as a result of the new observations or new data

Popper argued that it is not possible to prove beyond all doubt that a theory is 'true', but what is possible is that a theory can be proved as false. New evidence can arise, be discovered or produced that can prove a theory is false. Popper further argued that inductive reasoning is flawed because we cannot claim to know 'truth' from what invariably is limited observation. Just because we observe many instances of a phenomenon or behaviour does not mean that we can generalize to all instances of that phenomenon or behaviour. The famous example cited by Popper to exemplify this is that of the 'black swan'. An observer could record many thousands of instances of white swans and generalize that 'all swans are white'. However, it is not possible to make this conclusion, since not all swans have been observed. If one person subsequently observed a black swan, then the theory that all swans are white can be declared false.

Popper's view was that 'good' science should be based around the idea of falsifying theories rather than attempting to prove theories. The researcher should make it clear in their findings the conditions under which the theory being proposed can be falsified. This approach means that researchers who seek to defend a theory and find ways of proving it is true, even in the face of evidence to the contrary, are not adhering to the fundamentals of scientific method. A general principle behind scientific method, therefore, is that it should be possible to falsify a theory, and 'good science' should be centred on this and not on seeking to prove existing theories.

This debate is particularly relevant to economics because of the criticism that the subject has faced, particularly since the Financial Crisis of 2007–9. You may well find that some of your lecturers are fierce defendants of particular theories, or at the very least will seek to find ways in which a theory, or elements of a theory, can be adjusted or explained in the light of evidence which may suggest the theory is false. As you continue your studies, it is important to keep in mind the discussion in this section about the way we discover and verify new theories, knowledge and understanding about the subject.

Economics is a dynamic subject and the detailed research that many economists continue to undertake might hide the depth to which many take the criticisms of the subject. Popular criticism of the subject

can be just that – an attempt to appeal to the masses, many of whom will have little or no understanding of exactly what economists do, how they research into the subject or the controls they put in place to improve the quality of the research and outcomes. It is worth bearing in mind that the modelling method allows economists to approach issues, research them, think about them and attempt to uncover new knowledge and understanding. This is not the same thing as saying the models themselves, whatever they be, are final truths and are more important than the method.

While this book will include many theories which have been the focus of some criticism, they are important in understanding the historical development of economics and how we have come to know what we know. However, it is also important to recognize that there is still much we do not know and much to be discovered. Every economist knows this.

Keeping an Open Mind Understanding the processes and debates around scientific method, understanding the theories and their limitations helps generate more questions and the search for better ways in which we can understand the economy and human behaviour. All we ask is that you keep an open mind and recognize that there can be some truths hidden in theories, even if they are not the full truths, and these theories and the discipline as a whole are subject to ongoing evolution. Economics is *a way of approaching problems and issues* rather than a set of definitive truths. The debate about how economists come to know things and present theories and models which claim to be predictive is one which continues to pervade the discipline.

The Cambridge economist, Joan Robinson, perhaps captured the debate very well when she wrote economics ‘limps along with one foot in untested hypotheses and the other in untestable slogans ... our task is to sort out as best we may, this mixture of ideology and science’ (Robinson, J. (1968) *Economic Philosophy*. Pelican).

As noted earlier, separating out cause and effect can be problematic. Observation and experience can lead to the identification of phenomena occurring which intuition would seem to suggest are related in some way. Empirical research can help provide a conclusion which provides an answer, for example, whether a rise in the money supply does cause a rise in the price level. The question which must be asked is, ‘How do we know this “answer” is correct?’ What are the factors that influence the price level? How significant is the role of the money supply in determining the price level? How was the research conducted, and what ‘facts’ and assumptions were used in building the model? Can these facts and assumptions be accepted as an accurate representation of the ‘truth’, or are there interpretations of both which might impact on the conclusions drawn?

If facts and assumptions are accepted, then we must presume that those who collected them did so in an unbiased and unprejudiced way and that they were professionally competent and had sufficient expertise to be able to do so in a way we can trust. Separating out cause and effect can be informed by statistical tests but is also subject to interpretation. It is not always easy to establish cause and effect, particularly when controlled experiments are not possible, and this characterizes much of economics.

The Role of Assumptions

If you ask a physicist how long it would take for a cannonball to fall from the top of the Leaning Tower of Pisa, they will probably answer the question by assuming that the cannonball falls in a vacuum. This assumption is false; the building is surrounded by air, which exerts friction on the falling cannonball and slows it down. Yet the physicist will point out that friction on the cannonball is so small in relation to its weight that its effect is negligible. Assuming the cannonball falls in a vacuum greatly simplifies the problem without substantially affecting the answer.

Economists make assumptions for the same reason: assumptions can simplify the complex world and make it easier to understand. To study the effects of international trade, for example, we may assume that the world consists of only two countries and that each country produces only two goods. The real world consists of dozens of countries, each of which produces thousands of different types of goods, but by assuming two countries and two goods, we can focus our thinking. Once we understand international trade in an imaginary world with two countries and two goods, we are in a better position to understand international trade in the more complex world in which we live.

The art in scientific thinking is deciding which assumptions to make. Suppose, for instance, that we were dropping a beach ball rather than a cannonball from the top of the building. Our physicist would realize that the assumption of no friction is far less accurate in this case: friction exerts a greater force on a beach ball than on a cannonball because a beach ball is much larger and, moreover, the effects of air friction may not be negligible relative to the weight of the ball, because it is so light. The assumption that gravity works in a vacuum may, therefore, be reasonable for studying a falling cannonball but not for studying a falling beach ball.

Similarly, economists use different assumptions to answer different questions. Most economic issues will be affected by a number of different factors. If we try to model the issue taking into account all these factors, the complexity might lead to outcomes which do not help in developing an understanding of economic phenomena. In researching a phenomenon, economists will look at what happens when one factor changes, but all other factors assumed to have an effect are held constant. This is a core feature of neo-classical economic methodology. It might be assumed that the amount consumers wish to purchase is affected by the price of the good concerned, income, tastes and the prices of other related goods. Our understanding of consumer behaviour is simplified if we look at the effect on demand of a change in income and hold all other factors constant. This can be repeated with the other factors to generate some general principles about the demand for goods and services.

Assumptions must be tested to see the extent to which they are accurate and reasonable in the same way that it is deemed reasonable by the physicist to drop the assumption of friction when considering the effect of dropping a cannonball from the Leaning Tower of Pisa.

SCHOOLS OF THOUGHT

Given our preceding discussion on economic methodology, it might come as no surprise that there are different approaches to economics, and different perspectives.

These may be informed by assumptions and belief systems which influence the way issues are looked at, and the outcomes and policy implications which arise as a result. Perhaps the dominant methodology is the *neo-classical* approach, which is sometimes referred to as 'mainstream economics'.

Neo-classical Economics

The neo-classical approach takes the view that the market is a central feature in generating well-being and in answering the three questions all societies have to face, which we looked at in Chapter 1. In analyzing markets and outcomes, the neo-classical approach assumes that decisions are based on rationality, that economic agents act out of self-interest, and are autonomous. The neo-classical approach models behaviour through constrained optimization problems. This means that it is assumed economic agents seek to maximize or minimize outcomes but are subject to constraints. Individuals seek to maximize utility subject to the constraint of their income; firms seek to minimize costs subject to the constraint of resources available and the price of those resources.

Critics of this approach argue that the assumptions are flawed, and that what is observed about human behaviour does not conform to the neo-classical view. They argue that such is the power of the neo-classical hold on economics that other views, so-called heterodox economics (where the term 'heterodox' means views at odds with the mainstream), find it hard to gain any ground. These differing views include feminist economics, Marxist economics and the Austrian school.

Feminist Economics

Feminist economics questions many of the assumptions of the neo-classical school. Economic well-being, they argue, is not simply provided through market exchange but also includes unpaid work carried out in the home. This housework, by both males and females, needs to have the recognition its significance to well-being deserves. Economic activity, therefore, needs to include a valuation of this unpaid work.

Feminist economists also research into other areas where there are gender and social inequalities, and they would argue that it is not possible to have value-free analysis and research into economic issues. For example, the idea that humans face a trade-off between work and leisure is misleading in that 'leisure' is associated with pleasurable activities which people choose to take part in. For many women, 'non-work' activity, i.e. that which is not paid, is not leisure at all and involves considerable work in caring for the home and family. To only assume that 'work' is valuable betrays a value judgement which relegates analysis of unpaid work below that of paid work.

Marxist Economics

In later chapters, we will look at the working of markets and firms in more detail. Much of the analysis will be derived from the neo-classical approach, but there are other explanations for how markets and firms work. Marxist economics presents different explanations for essentially the same phenomena and has developed from the work of Karl Marx in the nineteenth century. Marx sought to analyze and understand the capitalist system and explain how and why production takes place and the circumstances under which different groups in society have economic power.

Marxist economics views firms and markets not as entities but as collections of humans, and it is these humans who make decisions. Some humans have control over the means of production and can exploit that power in ways which lead to different outcomes and which drive dynamism in economies. This dynamism can be self-destructive, however, and the competition between capitalists to attempt to retain control over the means of production is partly what generates booms and busts in capitalist economies. Neo-classical economists propose different explanations for the swings in the business cycle.

The Austrian School

The Austrian school originated in work carried out at the University of Vienna in the latter part of the nineteenth century. Academics at Vienna were of the belief that economic well-being is maximized when markets are allowed to do their work and that the government should have a minimal role in the economy (referred to as 'laissez-faire' roughly translated as 'to leave', or 'let it be'). Individual liberty is a fundamental principle in Austrian school economics.

The Austrian school is now not based in Vienna but has adherents in different parts of the world. Key figures in the Austrian school include Carl Menger, Eugen Böhm-Bawerk, Friedrich Weiser, Ludwig von Mises and Friedrich August von Hayek. Other influential economists such as the 1991 Nobel Prize winner in Economics, Ronald Coase, were said to have been influenced by the Austrian school (Coase was at the London School of Economics when Hayek was on the faculty and acknowledged the impact he made).

Austrian school economists look at the explanation for business cycles in the supply side of the economy rather than focusing on demand. Excess supply is what drives the economy into recession and this, in turn, can be caused by interest rates being too low, leading to too much investment and the availability of cheap money. It is this that triggers inflation. For Austrian economists, therefore, inflation is not the main problem or focus of policy; inflation is a symptom of imbalances in the financial sector of the economy. Economists from the Austrian school had been warning of too low interest rates and too high debt levels for many years in the early part of the twenty-first century and there are those who argue that it was these economists who correctly predicted the Financial Crisis of 2007–9 and not mainstream economists. Critics of the Austrian school argue that it relies on narrative analysis rather than mathematical, statistical and empirical analysis and so their claims cannot be tested and verified.

THE ECONOMIST AS POLICY ADVISOR

Often economists are asked to explain the causes of economic events and recommend policies to improve economic outcomes. Why, for example, is unemployment higher for teenagers than older workers, and given this situation what should the government do to improve the economic well-being of teenagers? These two roles lead to important distinctions in the way in which we need to view statements and

analysis. To answer the first question, the economist might use scientific method to offer an explanation, but the second involves a value judgement. This highlights the distinction between what is termed positive and normative economics.

Positive versus Normative Analysis

Suppose that two people are discussing minimum wage laws.

Pascale: Minimum wage laws cause unemployment.

Sophie: The government should raise the minimum wage.

Pascale's statement is making a claim about how the world works. Sophie is making a value judgement about a change she would like to see implemented.

Pascale's statement is referred to as a positive statement. **Positive statements** have the property that the claims in them can be tested and confirmed, refuted or shown to not be provable either way. It would be possible to conduct research to show whether there is any correlation between the imposition of minimum wage laws and a rise in unemployment. A positive statement does not have to be true – it is possible that the research might conclude that there is no link between minimum wages and unemployment.

positive statements claims that attempt to describe the world as it is

Sophie's is said to be normative. **Normative statements** have the property that they include opinion and make a claim about how the world ought to be; it is not possible to test opinions and confirm or reject them.

normative statements claims that attempt to prescribe how the world should be

Positive analysis incorporates the use of scientific methodology to arrive at conclusions which can be tested. Normative analysis is the process of making recommendations about particular policies or courses of action. It is perfectly possible to conduct both positive and normative analysis. For example, the statement: *the government should reduce the deficit as this will benefit the economy*, contains a normative statement – an opinion that the government ought to reduce the deficit. It also includes a positive statement: *A reduction in the government deficit will benefit the economy*, which is capable of being tested.

A key difference between positive and normative statements, therefore, is how we judge their validity. Deciding what is good or bad policy is not merely a matter of science; it also involves our views on ethics, religion and political philosophy.

Of course, positive and normative statements may be related. Our positive views about how the world works affect our normative views about what policies are desirable. Pascale's claim that the minimum wage causes unemployment, if true, might lead us to reject Sophie's conclusion that the government should raise the minimum wage.

WHY ECONOMISTS DISAGREE

If economics is classed as a science and adheres to scientific methods, why does there appear to be considerable disagreement among economists surrounding many different policy initiatives? There are two basic reasons:

- Economists may disagree about the validity of alternative positive theories about how the world works.
- Economists may have different values and, therefore, different normative views about what policy should try to accomplish.

Let's discuss each of these reasons.

Differences in Scientific Judgements

History shows us that there have always been disagreements between scientists about 'truth' and reality. In 1964, for example, Peter Higgs at the University of Edinburgh had his original paper on the theoretical model predicting what came to be known as the Higgs Boson, rejected by the journal *Physics Letters*, which saw the theory as having 'little relevance to physics'. In 2012, the experiments at Cern in Switzerland confirmed the existence of the Higgs Boson, and in 2013, Professor Higgs was jointly awarded the Nobel Prize for Physics. Science is a search for understanding about the world around us. It is not surprising that as the search continues, scientists can disagree about the direction in which truth lies.

Economists often disagree for the same reason. Economics is a young science, and there is still much to be learned. Indeed, there are some who argue that economics can never be a true 'science' because processes that are considered appropriate and necessary in natural sciences cannot be applied to economics, because it deals with human behaviour. Humans cannot be subjected to the same controls and comparisons that can be used in physics, for example.

Economists sometimes disagree because they have different beliefs about the validity of alternative theories or about the size of important parameters. For example, economists disagree about whether the government should levy taxes based on a household's income or based on its consumption (spending). Advocates of a switch from an income tax to a consumption tax believe that the change would encourage households to save more, because income that is saved would not be taxed. Higher saving, in turn, would lead to more rapid growth in productivity and living standards. Advocates of an income tax system believe that household saving would not respond much to a change in the tax laws. These two groups of economists hold different normative views about the tax system because they have different positive views about the responsiveness of saving to tax incentives.

SELF TEST 'Sometimes, theories are worth defending. The experience of Peter Higgs is testament to this view.' Comment on this statement in relation to theories in economics and the principle of falsifiability.

Differences in Values

Anneka and Henrik both take water from the town well. To pay for maintaining the well, the town imposes a property tax on its residents. Anneka lives in a large house worth €2 million and pays a property tax of €10,000 a year. Henrik owns a small cottage worth €20,000 and pays a property tax of €1,000 a year.

Is this policy fair? If not, who pays too much and who pays too little? Would it be better to replace the tax based on the value of the property with a tax that was just a single payment from everyone living in the town (a poll tax) in return for using the well – say, €1,000 a year? After all, Anneka lives on her own and uses much less water than Henrik and the other four members of his family who live with him and use more water as a result. Would that be a fairer policy?

This raises two interesting questions in economics – how do we define words like 'fair' and 'unfair', and who holds the power to influence and make decisions? If the power is in the hands of certain groups in government or powerful businesses, policies may be adopted even if they are widely perceived as being 'unfair'.

What about replacing the property tax, not with a poll tax but with an income tax? Anneka has an income of €100,000 a year so that a 5 per cent income tax would present her with a tax bill of €5,000. Henrik, on the other hand, has an income of only €10,000 a year and so would pay only €500 a year in tax, and the members of his family who do not work don't pay any income tax. Does it matter whether Henrik's low income is due to his decision not to go to university, and take a low paid job? Would it matter if it were due to a physical disability? Does it matter whether Anneka's high income is due to a large inheritance from her family? What if it were due to her willingness to work long hours at a dreary job?

These are difficult questions on which people are likely to disagree. If the town hired two experts to study how the town should tax its residents to pay for the well, we should not be surprised if they offered conflicting advice.

This simple example shows why economists sometimes disagree about public policy. As we learned earlier in our discussion of normative and positive analysis, policies cannot be judged on scientific grounds alone. Economists give conflicting advice sometimes because they have different values.

SELF TEST Why might economic advisors to the government disagree about a question of policy such as reducing a budget deficit?

Decision-Making in Economics

It could be said that economics is the science of decision-making. The way that economists go about making or recommending decisions involves, first, identifying the problem or issue. For example, greenhouse gas emissions are a contributory factor in climate change. Note that this is a premise which is assumed to be 'true'. One answer to this problem is a decision to cut the emissions of greenhouse gases. The next stage is to look at the costs and benefits involved in the decision. These costs and benefits are not just the private costs and benefits to individuals, firms and organizations; they will also include the costs and benefits to third parties who are not directly involved in the actual decision. For example, cutting greenhouse gas emissions means that resources will have to be diverted to new ways of production or different ways of producing energy. The private costs will be those borne by the businesses that will have to implement measures to adhere to the limits placed upon them. The social costs might include the impact on local people of the construction of wind farms or new nuclear power stations.

Having identified the costs and benefits, the economist then seeks to place a value on them to get some idea of the relationship between the costs and benefits of making the decision. In some cases, valuing costs and benefits can be easy, but many are much more challenging. The loss of visual amenity for a resident living near a wind turbine or the value of the possible loss of life from a nuclear catastrophe at a power plant, for example, may be very difficult to value. Economists have attempted to devise ways in which these values can be estimated, but they are not perfect.

Once the sum of the costs and benefits is calculated, the decision then becomes clearer. If the cost outweighs the benefit then making the decision may be unwise, but if the costs are less than the benefits, then it may mean the decision can be supported. Policymakers may want to look at the *extent* to which the costs outweigh the benefit, or the benefit outweighs the costs. Every day millions of decisions are made by individuals, businesses and governments. While not every one of these decisions will be made using the exact processes outlined above, and many of us certainly do not stop to think about how we rationalize our decisions, nevertheless our brains do engage in computational processes as we make decisions, but they are mostly subconscious. Economists and psychologists are increasingly finding out more about how humans make decisions, which is helping improve our understanding of the models which we use to analyze consumer behaviour.

SUMMARY

- Economics is characterized by different methodologies and approaches, including neo-classical, feminist, Marxist and Austrian.
- There is a debate about whether economics is a 'science'. It does follow certain scientific methodologies, but it must be accepted that economists are working with human behaviour.
- Economists make assumptions and build simplified models to understand the world around them. Economists use empirical methods to develop and test hypotheses.
- Economists must try to distinguish between cause and effect, and this is not always easy to do.
- Research can be conducted by using inductive and deductive reasoning – no one way is the 'right way'.
- Economists develop theories which can be used to explain phenomena and make predictions.
- The principle of falsifiability is based on the assumption that we cannot know everything for sure and, as a result, researchers should clarify the conditions under which a theory can be proved false.

- Using theory and observation is part of scientific method, but economists must always remember that they are studying human beings and humans do not always behave in consistent or rational ways.
- Economists can conduct experiments in laboratory settings and use ‘natural experiments’ observing outcomes from changes in policy or when events occur.
- A positive statement is an assertion about how the world *is*. A normative statement is an assertion about how the world *ought to be*.
- Economists who advise policymakers offer conflicting advice either because of differences in scientific judgements or because of differences in values. At other times, economists are united in the advice they offer, but policymakers may choose to ignore it.
- Decision-making in economics can be made more informed by assessing the costs and benefits of a decision and attempting to quantify the costs and benefits to provide the basis for an informed decision.

IN THE NEWS



The State of Economics

You do not have to look too far to find plenty of debate about the state of economics. There are books written which are heavily critical of mainstream economics, including ‘a chilling tale’ by John Quiggin called *Zombie Economics: How Dead Ideas Still Walk Among Us*, Ken Blawatt’s *Marconomics*, and Rod Hill and Tony Myatt’s *The Economics Anti-Textbook* to name but three.

Online there are also plenty of articles and blogs which extend the debate. One such example is Cahal Moran’s ‘Why the Problem Is Economics Not Economists’, which appeared on the Open Democracy *New Thinking for the British Economy* website. Moran does note that many economists are very frustrated with what seems to be a conventional wisdom that all of economics is rooted in neo-classical methodology, and that they are wedded to free markets oblivious to any limitations or weaknesses in the models they use. Economists who do economics know better, and Moran cites two of his colleagues at the University of Manchester, Rachel Griffiths and Diane Coyle (who is now at the University of Oxford), as two examples of such researchers.

For other economists, the issue about whether economics is a science is ‘sterile and crushingly boring’, as noted by Kartik B. Athreya in his book *Big Ideas in Macroeconomics: A Nontechnical View*. Those like Athreya who spend their days doing economics, know of the limitations of the subject and are well versed in scientific method and process.

This does not, however, seem to stem the flow of criticism of the subject. Hill and Myatt, for example, note ‘The typical introductory economics textbook teaches that economics is a value-free science that economists have an agreed upon methodology; and they know which models are best to apply to any given problem ... This *Anti-Textbook* points out that all this is a myth’ (p1). Blawatt’s opening section is titled ‘The Flagging World of Mainstream Classical Economics’, and Chapter 1 is titled ‘Economics of Power: Failure of Classical Economics’. The sleeve jacket of Quiggin’s book notes: ‘Zombie economics takes the



If you choose to read anything in economics, including criticisms of the subject, it is important to do so with a critical eye.

reader through the origins, consequences and implosion of a system of ideas whose time has come and gone.' If you choose to read anything in economics, including criticisms of the subject, it is important to do so with a critical eye. In the examples given above, for example, how many of the statements made are positive and how many normative?

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Critical Thinking Questions

- 1 Why do you think that economics has commanded such a barrage of criticism, particularly since 2008?
- 2 Write a defence of economic models as a methodology for finding out new information about economics.
- 3 Now write a criticism of economic models as a methodology for finding out new information about economics. Which of your arguments do you find most convincing and why?
- 4 Choose one of the references cited. If you could ask the author or authors of the book or article two questions about their book or article and the argument they are promoting, what would they be and why?
- 5 Look at the quotes provided in the last paragraph of the article. Try to identify which of the statements made are positive statements and which are normative and give a reason for your judgement in each case.

QUESTIONS FOR REVIEW

- 1 How is economics like a science?
- 2 Why do economists make assumptions?
- 3 Should an economic model describe reality exactly?
- 4 What is meant by empirical study in economics?
- 5 Using an example, explain the difference between inductive and deductive reasoning.
- 6 Should economic theories be developed as a result of observation or before observation? Explain.
- 7 What is the difference between a positive and a normative statement? Give an example of each.
- 8 Why do differences in values lead to disagreements among economists?
- 9 Using an example, explain the difference between an endogenous and an exogenous variable.
- 10 Why do economists sometimes offer conflicting advice to policymakers?

PROBLEMS AND APPLICATIONS

- 1 Terms like 'investment', 'capital', 'interest', 'price' and 'cost' have different meanings in economics than they do in normal everyday usage. Find out what the differences are and explain why economists might have developed these different meanings.

- 2 One common assumption in economics is that the products of different firms in the same industry are indistinguishable. For each of the following industries, discuss whether this is a reasonable assumption:
- steel
 - novels
 - wheat
 - fast food
 - mobile phones (think carefully about this one)
 - hairdressers.
- 3 A researcher in a university notices that the price of flights to holiday destinations tends to be much higher outside semester dates. They formulate a theory to explain this phenomenon. Has the researcher arrived at the theory by induction or deduction? What steps might the researcher take to apply scientific method to test the theory?
- 4 A politician makes a speech which is critical of the government's immigration policy, saying that it is too loose and encourages too many people to enter the country and take jobs away from local people. How might an economist go about assessing the validity of the politician's comments?
- 5 If models are not capable of representing the real world in any detail and rely too much on assumptions, then what value can they be?
- 6 Does the fact that there are different schools of thought in economics reduce its validity as an academic discipline?
- 7 Rival political groups argue about the value and effectiveness of speed cameras as a means of influencing driver behaviour and improving safety on the roads. An economist is asked to conduct research into the costs and benefits of speed cameras to help decision-making. What sort of factors will the economist have to take into consideration in such research, and what might be the challenges in identifying and quantifying the full range of costs and benefits?
- 8 If you were prime minister, would you be more interested in your economic advisors' positive views or their normative views? Why?
- 9 Would you expect economists to disagree less about public policy as time goes on? Why or why not? Can their differences be completely eliminated? Why or why not?
- 10 Consider a theory which states that an increase in interest rates will lead to an increase in savings. How would the principle of *ceteris paribus* be important in investigating the predictive power of this theory?